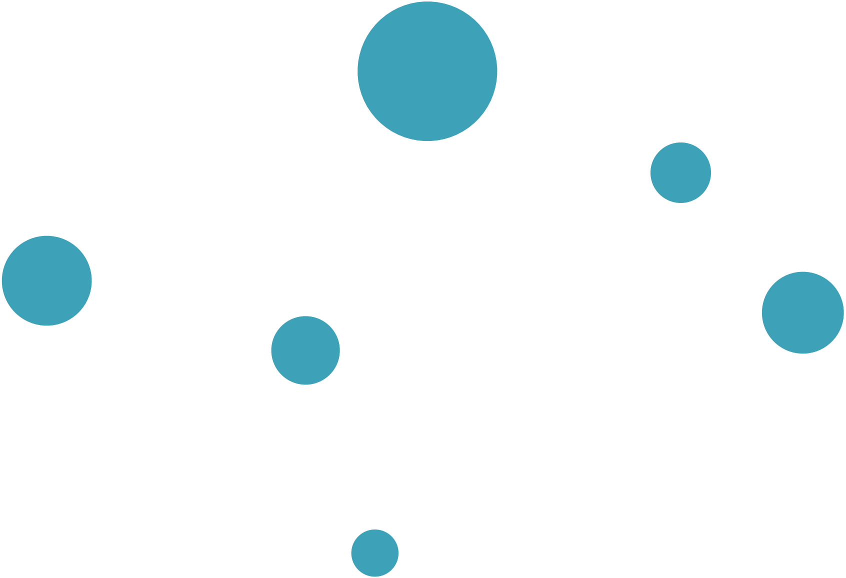
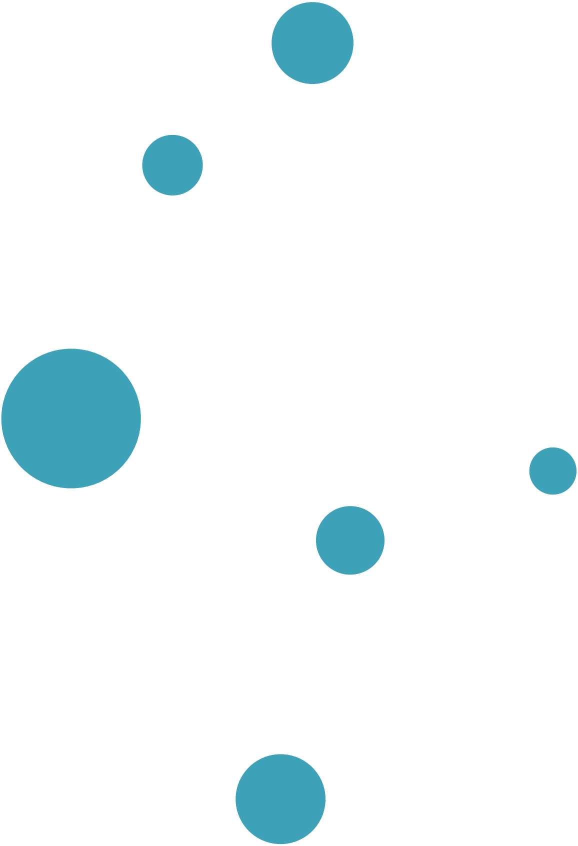
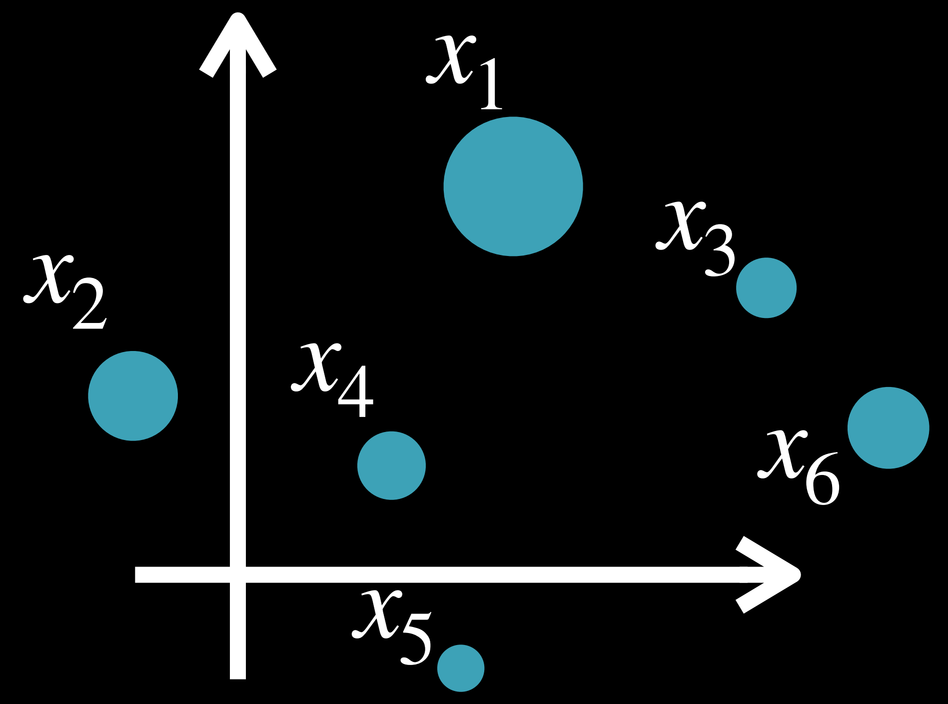


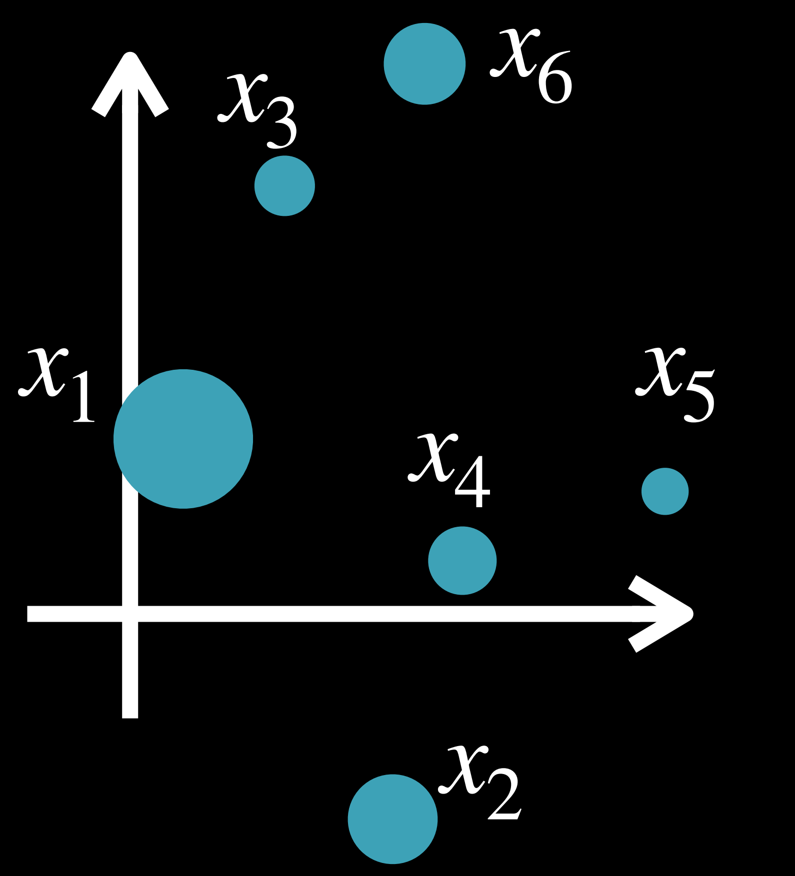


Pop-Quiz

What is the push-forward of a discrete probability measure $\alpha = \sum_{1}^{n} \mathbf{a}_i \delta_{x_i}$?



$\Longrightarrow$



$$\alpha = \sum_{i=1}^{n} \mathbf{a}_i \delta_{x_i}$$

$$f_\sharp \alpha = \sum_{i=1}^{n} \mathbf{a}_i \delta_{f(x_i)}$$

What is this f?

Re-visting Discrete Monge

Data indexed by
$i \in \{1,\ldots,n\}, j \in \{1,\ldots,n\}$

Cost matrix: $(C_{i,j})_{i,j=1}^{n} \in \mathbb{R}^{n \times n}$

Search over permutations in
$\Sigma_n = \{\sigma : \{1,\ldots,n\} \to \{1,\ldots,n\}\}$

We want the min-cost one

Monge's Problem:

(Matching/permutation version)

$$\min_{\sigma \in \Sigma_n} \frac{1}{n} \sum_{i=1}^{n} \mathbf{C}_{i,\sigma(i)}$$